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Sentinel Lymph Node Biopsy in Clinically Node-Negative Merkel Cell Carcinoma: An Update of the Westmead Hospital Experience

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ABSTRACT

Background: Merkel cell carcinoma (MCC) is a rare, aggressive cutaneous neuroendocrine malignancy with a high risk of nodal metastasis. Sentinel lymph node biopsy (SLNB) is used to stage clinically node-negative disease and to guide adjuvant therapy decisions. This study updates the Westmead Hospital experience with an expanded cohort and extended follow-up.

Methods: Patients with clinically node-negative MCC who underwent SLNB between 1998 and 2024 were identified from an institutional database. Demographic, tumour, treatment and outcome data were reviewed.

Results: A total of 75 patients (57 men, 18 women; median age 74 years) underwent SLNB. A total of 28 (37%) had a positive SLNB and 47 (63%) a negative SLNB. Seven patients with a negative SLNB developed nodal recurrence, for a false-negative rate of 20%. Among patients with head and neck primaries, the false-negative rate was 12% (3/31). Overall recurrence occurred in 11 of 47 (23%) SLNB-negative and 11 of 28 (39%) SLNB-positive patients ($p=0.23$). Distant recurrence was more frequent after a positive SLNB (18% vs. 6%). Overall survival did not differ significantly by SLNB status ($p=0.11$).

Conclusion: MCC demonstrates a high rate of subclinical nodal metastases, with a SLNB positivity rate of nearly 40% in our series. These findings support the use of SLNB in the management of clinically node-negative MCC to improve staging accuracy, prognostication and guide management decisions, with an acceptable false-negative rate under close surveillance.

1 | Introduction

Merkel cell carcinoma (MCC) is a rare, aggressive primary cutaneous neuroendocrine malignancy arising predominantly in older adults, with Australian and New Zealand populations recording the highest incidence worldwide (approximately 2 per 100 000 per year) [1]. MCC has a higher disease-specific mortality than melanoma, with up to 30%–40% of patients dying from their disease [2]. Most patients (50%–60%) present with a primary lesion alone, and 20%–30% present with clinically involved lymph nodes. It demonstrates a strong propensity for nodal and distant metastases, though distant metastases at presentation are uncommon (< 5%) and usually occur as a relapse event [3, 4].

The presence of nodal metastases is an independent prognostic factor for poor outcomes. Patients with clinically node-negative but pathologically node-positive disease have significantly worse survival than those who remain node-negative [5]. Sentinel lymph node biopsy (SLNB) enables pathological staging of clinically node-negative (Stages I/II) MCC, identifying occult nodal disease in 20%–40% of cases and guiding adjuvant therapy decisions. However, false-negative results occur in approximately 10%–20% of cases, particularly in the head and neck region where lymphatic drainage is variable [3, 4, 6–8].

This study updates the Westmead Hospital experience of SLNB in MCC [9] with extended follow-up and an expanded cohort,

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providing a 27-year institutional perspective. The objectives were to determine SLNB positivity and false-negative rates; recurrence patterns according to SLNB status and evaluate the management implications of SLNB in clinically node-negative MCC.

2 | Methods

This study expands on our previously published institutional series of SLNB in MCC by including additional patients and extended follow-up [9].

Cases were identified from the institutional MCC database, which prospectively recorded cases from 1998 to 2021 by a single senior clinician using a data collection protocol. This was supplemented by a retrospective review of cases from 2021 to 2024 using institutional electronic medical records and cancer service databases to obtain the same dataset. Data were extracted from records by WSLHD staff as part of a quality assurance project and de-identified before analysis. This review was approved as a WSLHD quality assurance project.

Patients with clinically node-negative (Stages I/II) MCC who underwent SLNB at Westmead Hospital between January 1998 and December 2024 were eligible. Inclusion criteria were histologically confirmed MCC, absence of clinical or radiological nodal disease, SLNB performed, and at least 6 months of follow-up. Patients referred externally for ongoing management were excluded. Staging was according to the American Joint Committee on cancer's eighth edition Cancer Staging Manual [10]. Demographic, tumour and treatment variables were reviewed, including tumour site and size, SLNB findings, subsequent nodal treatment, recurrence and survival outcomes.

Standard routine surveillance following treatment included a 3-monthly review for the first 2 years, followed by a 6-monthly review until 5 years. Most patients also underwent FDG-PET imaging at approximately 3 months post-treatment, followed by cross-sectional imaging 6-monthly.

2.1 | Surgical Procedure

SLNB was performed as previously described [9] at the time of wide local excision of the primary MCC. Preoperative lymphoscintigraphy was performed the day before or on the day of surgery with four peritumoural, intradermal injections of technetium-99 sulphur colloid. The dual SLN localisation technique was used with intraoperative intradermal injection of 2 mL of patent blue dye (Laboratoire Guerbet, Aulnay-sous-Bois, France) in four quadrants around the primary tumour or biopsy scar. The SLN(s) were localised by gamma probe and visual confirmation

of patent blue dye. Each SLN was sent for histological examination, serially sectioned and stained with haematoxylin and eosin and subjected to immunohistochemical (IHC) analysis with anti-cytokeratin 20 and AE1/AE3 panels.

2.2 | Statistical Analysis

Descriptive statistics summarise patient and tumour characteristics. A false-negative SLNB was defined as recurrence within a nodal basin from which the original SLNB was negative, and we calculated the rate as false-negative SLNBs/(false-negative SLNBs + true-positive SLNBs), excluding true-negative cases. Associations between SLNB status and overall recurrence were analysed using chi-square tests. Tests were two-sided, with $p < 0.05$ considered statistically significant. Locoregional and distant recurrences were summarised descriptively. Given the modest cohort size and limited number of events, statistical analyses were descriptive and exploratory, intended to be hypothesis-generating. Analyses were performed with GraphPad Prism version 10.

3 | Results

A total of 75 patients with clinically node-negative MCC underwent SLNB, comprising 57 males and 18 females. Median age was 74 years, and median follow-up time was 39 months (range: 3–136 months). Four patients died within 8 months of treatment, resulting in a short follow-up duration. Median primary tumour size was 10 mm (range: 3–45 mm). By T stage, 64 patients (85%) had T1 disease and 11 (15%) had T2 disease. Most occurred in the head and neck or extremities. Eleven patients (15%) had a background of immunosuppression.

3.1 | SLNB Results

Details of recurrences are presented in Table 1. SLNB was positive in 28 patients (37%) and negative in 47 patients. Most patients underwent wide local excision of the primary tumour or re-excision of the initial biopsy site at the time of SLNB. Two patients underwent SLNB alone without excision of the primary lesion—one was located on the upper eyelid, and the other on the scalp with multiple surrounding satellite lesions. The baseline characteristics are outlined in Table 2.

All patients with a positive SLNB underwent nodal treatment, which consisted of radiotherapy ($n = 26$), completion lymphadenectomy (CLND) followed by adjuvant radiotherapy ($n = 1$), and CLND alone ($n = 1$) in a patient with prior long-course radiotherapy for rectal cancer.

TABLE 1 | SLNB status and site of recurrence.

SLNB status	<i>n</i>	Overall recurrence, <i>n</i> (%)	Locoregional (nodal), <i>n</i> (%)	Locoregional (local), <i>n</i> (%)	Distant recurrence, <i>n</i> (%)
Negative	47	11/47 (23)	7/11 (67)	1/11 (10)	3/11 (27)
Positive	28	11/28 (39)	1/11 (10)	5/11 (45)	5/11 (45)

TABLE 2 | Baseline characteristics of patients undergoing SLNB.

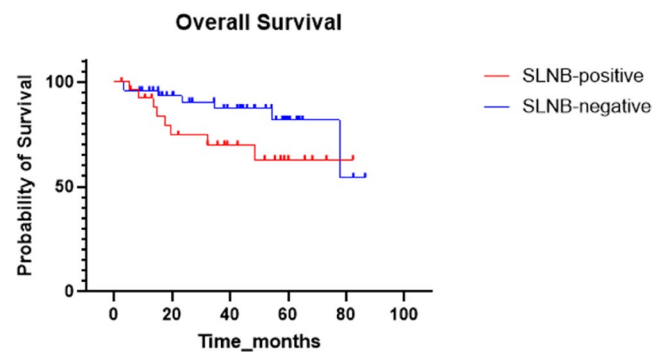
Characteristic (<i>n</i> = 75)	All patients, <i>n</i> (%)	SLNB negative, <i>n</i> (%)	SLNB positive, <i>n</i> (%)
Number of patients	75	47 (63)	28 (37)
Median age at diagnosis		74	
Male	57 (76)	34 (72)	23 (82)
Female	18 (24)	13 (28)	5 (18)
Immunosuppression	11 (15)	9 (19)	2 (7)
Location of the primary tumour			
Head/neck	31 (41)	22 (47)	9 (32)
Upper limb	20 (27)	15 (32)	5 (18)
Lower limb	17 (23)	7 (15)	10 (36)
Trunk	7 (9)	3 (6)	4 (14)
T stage primary lesion			
T1 (< 1 cm)	27 (36)	19 (40)	8 (29)
T1 (1–2 cm)	37 (49)	24 (51)	13 (46)
T2 (> 2 cm)	11 (15)	4 (9)	7 (25)

Following a negative SLNB, 46 patients (98%) did not undergo further nodal treatment and were observed. One patient with a delay in commencing adjuvant radiotherapy to the primary site developed an early clinical nodal disease recurrence, which was treated with radiotherapy.

3.2 | Recurrence

Overall, recurrence occurred in 11 of 47 (23%) SLNB-negative patients and 11 of 28 (39%) SLNB-positive patients. The odds of any recurrence were approximately doubled in SLNB-positive patients (OR: 2.12; 95% CI: 0.77–5.84), but this difference was not statistically significant ($\chi^2 = 1.44$, $p = 0.23$).

Six patients developed a locoregional recurrence in the SLNB-positive group. One patient developed nodal recurrence following CLND alone. Three patients received adjuvant radiotherapy to the primary site and radiotherapy alone to the lymph node basin and developed a locoregional dermal recurrence outside the radiotherapy field. Two patients received radiotherapy alone to the lymph node basin, without radiotherapy to the primary site and developed a local recurrence. One patient had undergone a re-excision of the primary lesion on the lower leg with no residual disease identified, followed by a split skin graft. The recovery was complicated by subsequent wound infection on a background of diabetes and obesity. The other patient had the primary lesion excised from the dorsum of the hand with clear margins, followed by a full-thickness skin graft repair. Eight patients developed a locoregional recurrence in the SLNB-negative group, of which seven were nodal recurrences and one was a dermal recurrence adjacent to the treated primary site. None of the patients in the SLNB-negative group who developed

**FIGURE 1** | Overall survival of SLNB-positive and SLNB-negative patients.

nodal recurrence had undergone further treatment of the relevant nodal basin following the SLNB. The false-negative rate of SLNB was 20% (seven patients with a nodal recurrence). The negative predictive value was 85% (40 of 47 SLNB-negative patients). Of the seven false-negative cases, three involved the axilla, three the head and neck region and one the groin. All recurrences occurred within the original nodal basin. Of these patients, six underwent adjuvant radiotherapy to the primary site. Amongst patients with a nodal recurrence following a negative SLNB, the median time to nodal relapse was 9 months (range: 3–14 months).

Distant recurrence occurred in three patients in the SLNB-negative group and five patients in the SLNB-positive group. Among SLNB-positive patients who developed distant disease, four had received radiotherapy alone to the lymph node basin and one had undergone both CLND and radiotherapy to the nodal basin.

3.3 | Status

At the time of last follow-up, 53 patients had no evidence of disease, 7 were alive with disease and 4 had died of other causes. In total, 11 patients had died of MCC: 8 with a positive SLNB and 3 with a negative SLNB.

Kaplan–Meier analysis (Figure 1) demonstrated no statistically significant difference in overall survival between SLNB-positive and SLNB-negative patients (log rank $p = 0.11$). The hazard of death was higher in SLNB-positive patients (HR = 2.24, 95% CI: 0.77–6.54), but this did not reach statistical significance. Median overall survival was not reached in either group. These survival analyses were exploratory and descriptive, reflecting the small cohort and limited number of events.

4 | Discussion

SLNB has an important role in the management of clinically node-negative MCC. It provides accurate pathological staging, guides decisions for adjuvant therapy and provides valuable prognostic information.

The American Joint Committee on Cancer Eighth Edition Cancer Staging Manual [10] established SLNB-positive patients

as Stage IIIA, compared to clinically node-positive patients as Stage IIIB, based on the better prognosis associated with the early detection of subclinical nodal metastases. The 2024 National Comprehensive Cancer Network (NCCN) [11] guidelines and ESMO-EURACAN Clinical Practice Guideline [12] recommend SLNB should be performed where possible before the excision of the primary tumour. Both guidelines recommend observation as the preferred option in SLNB-negative patients, with the consideration of radiotherapy in patients at high risk for a false-negative.

Accordingly, at our institution we perform SLNB in all suitable patients to pathologically identify and stage the nodal basin with the intent of avoiding nodal treatment in SLNB-negative patients. This study extends our prior institutional experience with longer follow-up and additional patients, providing a 27-year perspective on recurrence patterns and false-negative rates [9]. The Westmead Hospital experience represents, to date, the largest series of Australian patients undergoing SLNB in clinically node-negative MCC. In this updated cohort, the SLNB-positive rate was 37% and the overall false-negative rate was 20%. For patients with primary tumours located in the head and neck, the SLNB-positive rate was 29% and the false-negative rate 12% (3 of 31 patients in total). These figures are consistent with the international literature, which reports SLNB positivity in approximately 20%–40% of patients and false-negative rates of 10%–20% [3, 4, 6–8].

In our cohort, overall recurrence was approximately double in SLNB-positive compared with SLNB-negative patients (39% vs. 23%), though this was not statistically significant. Distant recurrence was higher in the SLNB-positive patient group (18% vs. 6%). This is expected and consistent with the review completed by Gunaratne et al. [3], which found distant relapse was more common following a positive rather than a negative SLNB (17.6% vs. 7.3%, $p < 0.001$). We did not find a significant survival difference between the SLNB-positive and SLNB-negative groups. Though our sample size was small, it is possible that the treatment approach of radiotherapy for SLNB-positive patients may have mitigated the risk of the expected survival difference.

The prognostic value of SLNB in MCC is supported by the literature. Positive lymph node status, even in early disease, is associated with poorer outcomes and survival [13, 14]. The Mayo Clinic series of 150 patients [8] found that patients with a positive SLNB had almost double the incidence of regional and distant recurrences, with a statistically significant higher incidence of in-transit recurrences. Analysis of 5823 patients from the National Cancer Database by Lemos et al. demonstrated improved 5-year overall survival in patients with a negative SLNB compared with those who did not undergo pathological nodal evaluation (76% vs. 59%, $p < 0.001$) [15]. Similarly, Kachare et al. [5] analysed SEER registry data and reported superior 5-year MCC-specific survival for SLNB-negative patients (84.5% vs. 64.6%, $p < 0.001$), as well as improved survival among those undergoing SLNB compared with observation alone (79.2% vs. 73.8%, $p = 0.004$). Another large retrospective study by Dugan et al. [16] also found that patients who did not undergo SLNB had significantly worse recurrence-free survival, disease-specific survival and overall survival ($p < 0.0001$) than those who underwent SLNB.

In another large single-institution study of 153 patients, Fields et al. [4] identified lymphovascular invasion in the primary tumour to be associated with SLNB positivity, recurrence and MCC-specific death. However, in contrast, they did not demonstrate a significant survival difference between SLNB-positive and SLNB-negative patients. Histopathological features have emerging prognostic value. A study by Zijlker et al. [17] investigated histopathological prognostic factors and found that patients with a lower SLN tumour burden (< 0.625 mm) had a significantly better 2-year overall survival than those with a higher tumour burden (90% vs. 63%, $p = 0.046$). This shows similarities to melanoma, where nodal tumour burden is factored into staging. In multivariate analysis, immunosuppression and extracapsular extension were also significant prognostic factors for overall survival. Due to small numbers and missing data, specific histopathological features such as lymphovascular invasion and SLN burden and their relationship with overall survival were not assessed in our study.

The meta-analysis by Kanakopoulos et al. [7] included 17 studies with over 22 000 patients who were eligible for or underwent SLNB. It found that undergoing an SLNB was associated with an increased likelihood of adjuvant postoperative treatments, including chemotherapy, radiotherapy and completion lymph node dissection, demonstrating the influence nodal status has on adjuvant treatments. However, recurrence rates did not significantly differ between those who underwent SLNB and those who did not, although only four studies provided direct comparisons. Similarly, Tarantola et al. [18] in a cohort of 240 patients, found no significant difference in regional nodal recurrence and overall survival in those who did and did not undergo pathological nodal staging with either an SLNB or an elective lymph node dissection. Despite this, it remains likely that SLNB has an important role in risk stratification and guiding adjuvant therapy selection.

Several factors can contribute to a false-negative SLNB, which have been included in the NCCN and ESMO guidelines [11, 12] as factors to be considered when determining the suitability of adjuvant treatment in the SLNB-negative patient; these include profound immunosuppression, the risk of anatomic compromise due to preceding surgery, and the likelihood of aberrant lymph node drainage and the presence of multiple SLN basins in the head, neck or midline trunk MCC. None of our seven patients with a false-negative SLNB had a background of immunosuppression. Patients with a mid-face or midline MCC may especially benefit from SLNB due to the uncertainty of the laterality of lymphatic drainage and the benefit of avoiding unnecessary nodal treatment if found to be negative. Our institution's low false-negative rate of 12% in patients with a head and neck primary supports this. However, a systematic review of SLNB in MCC of the head and neck [19] reported on a false-negative rate of 19.2% from a total of 136 patients. These findings highlight the importance of maintaining appropriate follow-up even in SLNB-negative patients, given the ongoing risk of nodal relapse despite a negative SLNB.

An alternative management paradigm is to forego SLNB and electively irradiate draining nodes with the assumption that many will contain subclinical nodal metastases. This approach is utilised by the Peter MacCallum Cancer Centre in Melbourne,

Australia [20], in clinically node-negative T2 MCC patients, performing SLNB only in T1N0 patients. This, however, commits all patients to regional treatment, including those who may have otherwise been appropriate for observation following a negative SLNB.

Our institution's approach of managing an SLNB-positive nodal basin with radiotherapy is supported by the unique radio-responsiveness of MCC [21]. Also factoring into treatment decisions is the expected patient demographic of MCC, which is typically older with the risk of additional medical co-morbidities. Multiple retrospective series on SLNB-positive patients have demonstrated comparable outcomes for recurrence and survival with either definitive radiotherapy or CLND [22–24]. Fang et al. [22] found no difference in regional recurrence-free or disease-specific survival at 2 years between patients treated with definitive radiotherapy versus surgery ± adjuvant radiotherapy. Similarly, Perez et al. [23] reviewed 71 SLNB-positive patients treated with CLND, CLND + adjuvant radiotherapy or radiotherapy alone and observed regional recurrence rates of 0%, 5% and 7.5%, respectively, with no significant difference in disease-specific or overall survival. Lee et al. [24] likewise reported no difference in nodal or distant recurrence-free survival between CLND and RT in a cohort of 163 patients. These studies support radiotherapy as an effective, less invasive treatment option with treatment individualised according to medical co-morbidities, surgical risk and patient preferences.

Immunotherapy has been a major advance in the setting of metastatic MCC and is currently under investigation in the adjuvant setting for earlier-stage disease. Trials such as I-MAT (NCT04291885), STAMP (NCT03712605) and ADAM (NCT03271372) are investigating PD-1 and PD-L1 inhibitors following resection, based on early data suggesting potential improvements in disease-free survival [25]. In our cohort, four patients were treated on the double-blind I-MAT trial with either adjuvant avelumab or placebo. Three were SLNB-positive, and one was SLNB-negative; none had developed recurrence at last follow-up. As systemic options evolve, accurate staging with identification of subclinical nodal disease remains important for early identification of patients at a higher risk of recurrence for appropriate selection of treatment.

Limitations of this study include its modest size relative to larger centres and inherent selection bias. All patients had at least 9 months of follow-up, but there was variation in total follow-up duration for each patient, which may have impacted the validity of comparing recurrence rates between groups. However, it represents the largest Australian series of MCC patients undergoing SLNB at a single tertiary institution, with consistent surgical and pathological protocols. It contributes long-term Australian data to the literature and reinforces findings from prior institutional and pooled international analyses.

5 | Conclusion

MCC demonstrates a high rate of subclinical nodal metastases, with a SLNB positivity rate of nearly 40% in our series. These findings support the use of SLNB in the management

of clinically node-negative MCC to improve staging accuracy, prognostication and guidance management decisions. The risk of a false-negative SLNB is acceptable, provided there is close clinical and imaging surveillance for locoregional recurrence.

Author Contributions

Julie R. Howle: investigation, writing – review and editing, conceptualization. **Michael J. Veness:** conceptualization, supervision, writing – review and editing, methodology, investigation. **Julie Ahn:** conceptualization, data curation, formal analysis, writing – original draft, methodology, investigation, writing – review and editing.

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Disclosure

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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